

Supplemental Material:

Quantum-referenced refractive-index calibration of air at 1762 nm for environmental phase compensation in quantum optical interfaces

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I. CAMPAIGN TEMPORAL COVERAGE

The processed dataset spans 231 calendar days (2025-01-01 to 2025-08-20) but is not a continuous record: it consists of 104 observation segments, separated by gaps longer than 2 h, with 89 active observation days in total. The longest gap (111 days) separates a sparse January–February block from the main body of the campaign in June–August, which contains 91.7% of the observations and entirely exceeds the 25 °C validity bound of the Mathar model. The overall duty cycle, defined as the ratio of active segment time (35.6 days) to the calendar span, is 15.3%. Within active periods the median sampling interval is 12.9 s.

Table ?? summarises the monthly coverage. January and February are the only months within the Mathar validity domain ($T \in [10, 25]^\circ\text{C}$); February is represented by only 25 retained observations spread over 13 active days. Consequently the retained-observation mean conditions ($\bar{T} = 28.9^\circ\text{C}$, $\bar{\text{RH}} = 30.2\%$, $\bar{P} = 986.4\text{ hPa}$) are weighted toward the warm-season segments, and differ from the time-weighted means ($\bar{T}_{\text{tw}} = 25.6^\circ\text{C}$, $\bar{\text{RH}}_{\text{tw}} = 31.0\%$, $\bar{P}_{\text{tw}} = 987.9\text{ hPa}$) by -3.2°C in temperature. The regression coefficients of Table ?? are estimated from the retained observations, and the Kramers-Kronig evaluation of Sec. ?? uses the corresponding retained-observation mean conditions.

II. STATISTICAL VALIDATION OF THE ENVIRONMENTAL COEFFICIENTS

The refractive index time series exhibits serial correlation characteristic of slowly varying environmental parameters. The residual autocorrelation function decays to $1/e$ at a lag of approximately 78 samples, corresponding to a decorrelation time well resolved by the 10–20 s measurement cadence. To ensure that reported coefficient uncertainties account for this structure, we compute heteroskedasticity- and autocorrelation-consistent (HAC) standard errors with a Bartlett kernel [?] using a maximum lag of 156 samples, twice the measured $1/e$ decorrelation time. As the primary uncertainty estimate we use moving-block bootstrap resampling with a block length of 156 samples and $B = 5000$

replications [? ?]; the block length preserves the serial correlation structure of the original data while rendering successive blocks approximately independent. The HAC standard errors serve as an independent cross-check: the HAC and bootstrap standard errors agree to within 3% for all three coefficients (Table ??), confirming that the uncertainty estimates are robust to the specific autocorrelation-consistent estimator employed. For reference, a HAC estimate with an insufficient lag of 3 samples, which does not capture the dominant serial correlation, underestimates the standard errors by approximately a factor of five.

First-order residual autocorrelation is additionally addressed using the Prais-Winsten generalized least squares estimator [?]. The transformation improves the Durbin-Watson statistic from 1.197 (raw OLS residuals) to 2.226 (transformed residuals), where a value of 2.0 indicates no first-order autocorrelation [?]. The Prais-Winsten coefficient estimates agree with the OLS estimates within 1.5%, confirming that the point estimates are insensitive to the residual correlation structure.

The pressure coefficient reported in Table I of the main text ($+2.5949 \times 10^{-7} \text{ hPa}^{-1}$) follows from the regression coefficient ($+2.595 \times 10^{-9}$ in the native fit unit of Pa^{-1}) by conversion of the pressure unit from Pa to hPa (factor 10^2). The result agrees with first-principles Lorentz–Lorenz scaling of the air refractivity at standard conditions.

A. Uncertainty budget for single-epoch refractive index

Table ?? provides the uncertainty budget for the determination of the refractive index at a single measurement epoch. The dominant contribution is the residual scatter of the regression ($\delta n = 1.84 \times 10^{-7}$), which includes the spatial and temporal mismatch between the environmental sensor and the optical path as well as interferometric readout noise. A decomposition of this residual against a linear surrogate of the Mathar model evaluated on the identical data rows shows that the intrinsic non-linearity of the environmental response contributes only 4.4×10^{-8} , with the remaining 1.78×10^{-7} attributable to measurement noise. The 780 nm frequency reference contributes $\delta n = 1.0 \times 10^{-9}$; this follows from lock-

TABLE I. Monthly coverage of the processed dataset. n_{obs} : retained observations; active days: calendar days containing observations; cadence: median intra-month sampling interval; gap: longest intra-month gap; ranges are min–max within each month. The last column is the fraction of the month’s observations within the Mathar validity domain ($T \in [10, 25]^\circ\text{C}$).

Month	n_{obs}	Active days	Cadence (s)	Gap (d)	T range ($^\circ\text{C}$)	RH range (%)	P range (hPa)	In-domain
2025-01	12 109	17	13.4	4.5	17.5–21.4	28.5–40.9	964.6–1006.8	100%
2025-02	25	13	41 317	5.9	20.2–22.1	28.5–38.4	982.2–999.4	100%
2025-06	28 911	13	12.8	1.5	29.3–33.3	22.8–34.6	980.6–995.7	0%
2025-07	54 397	26	12.8	4.0	25.9–34.6	18.9–44.0	974.1–993.3	0%
2025-08	50 342	20	12.9	1.0	25.4–31.9	24.8–45.3	978.9–994.5	0%
Total		89	12.9	111.3	17.5–34.6	18.9–45.3	964.6–1006.8	8.3%

ing the 780 nm laser to the ^{87}Rb D₂ line via Doppler-free saturated absorption spectroscopy ($\delta\nu \approx 384\text{ kHz}$ at $\nu \approx 384.230\text{ THz}$), corroborated by long-term wavemeter monitoring of the laser frequency. The 1762 nm quantum frequency reference and interferometer non-linearity are negligible. The combined expanded uncertainty ($k = 2$) is 3.68×10^{-7} .

TABLE II. Uncertainty budget for single-epoch refractive index determination at 1762 nm.

Source	δn (1σ)
Residual scatter (incl. spatial/temporal mismatch)	1.84×10^{-7}
of which: measurement noise	1.78×10^{-7}
of which: model nonlinearity	4.4×10^{-8}
780 nm frequency reference	1.0×10^{-9}
1762 nm frequency reference	$< 10^{-12}$
Interferometer non-linearity	Negligible
Combined standard uncertainty	1.84×10^{-7}
Combined expanded uncertainty ($k = 2$)	3.68×10^{-7}

B. Uncertainty budget for environmental coefficients

Table ?? separates the statistical and systematic contributions to the uncertainty of the environmental coefficients. The statistical component is the moving-block bootstrap standard error ($B = 5000$, block length 156 samples), cross-checked against the HAC estimate. The systematic component accounts for the sensor gain error, that is, the uncertainty in the multiplicative scale of the sensor readings. A constant sensor offset is absorbed by the fitted intercept n_0 and does not affect the coefficients; the gain error, by contrast, propagates directly into the coefficient estimate as $\alpha_X^{\text{meas}} = \alpha_X^{\text{true}}/g_X$, where g_X is the sensor gain.

Worst-case gain bounds follow from the manufacturer specifications of the BME280 sensor over the observed campaign ranges (T : 17.5°C to 34.6°C , span 17.1 K ; H : 18.9% to 45.3% , span 26.4% ; P : 964.6 hPa to 1006.8 hPa , span 42.2 hPa). For temperature, the accuracy of 1.0°C bounds the gain error to $2 \times 1.0/17.1 = 11.7\%$. For humidity, the accuracy of 3% at 25°C degrades to about 4% over the temperature range of

the campaign (together with long-term drift), bounding the gain error to $2 \times 4.0/26.4 = 30.3\%$. For pressure, the accuracy of 1 hPa bounds the gain error to $2 \times 1.0/42.2 = 4.7\%$. These worst-case gain bounds translate directly into the endpoint sensitivities of Table ??: the change in the coefficient if the gain deviated from unity by the full bound, $\Delta\alpha_X^{\text{end}} = (\delta g/g)_{\text{max}} \cdot |\alpha_X|$.

For the standard systematic uncertainty u_{syst} we convert the worst-case gain bound into a standard deviation assuming a rectangular distribution of the gain over $[1 - \delta, 1 + \delta]$, i.e. division by $\sqrt{3}$. The same treatment is applied to all three channels. For the humidity channel the rectangular standard uncertainty is $3.99 \times 10^{-9}/\sqrt{3} = 2.30 \times 10^{-9}$. We also list, for comparison, the combined humidity difference significance evaluated in Sec. ?? using the paired bootstrap standard error of the difference rather than the single-coefficient standard error.

III. SAME-CONDITION COMPARISON WITH THE MATHAR MODEL

To compare the measured environmental response with the Mathar model on equal footing, we evaluate the Mathar formulation [?] at the identical (T, H, P) rows of the measurement campaign and fit the same linear surrogate model $n = n_0 + \alpha_T T + \alpha_H H + \alpha_P P$ to both the measured refractive index and the Mathar predictions. Both fits therefore share the same design matrix, and the coefficient difference is estimated with a paired moving-block bootstrap (identical block resampling for both models, $B = 5000$, block length 156 samples) that accounts for the correlation between the two estimates.

Table ?? summarizes the result for two analysis windows: the full campaign and the subset within the validity domain of the Mathar model ($T \in [10, 25]^\circ\text{C}$, comprising 12,134 of the 145,784 observations; 91.7% of the data lie above 25°C). For the full campaign, the measured α_T and α_P agree with the Mathar surrogate within 0.35% and 0.93% , respectively, and the humidity coefficient is smaller in magnitude than the surrogate by $+24.8\%$. Within the Mathar validity domain the humidity difference is not statistically significant. Because the Kramers-Kronig prediction at the mean campaign

TABLE III. Uncertainty budget for the environmental coefficients at 1762 nm. Statistical uncertainties are moving-block bootstrap standard errors ($B = 5000$, block length 156 samples); HAC (lag 156) standard errors agree within 3%. Systematic uncertainties are sensor gain bounds from BME280 manufacturers specifications converted to standard uncertainties assuming a rectangular gain distribution (division by $\sqrt{3}$); the endpoint sensitivity is the coefficient change for a gain deviation of the full worst-case bound.

Coefficient	Value	u_{stat}	Endpoint	u_{syst}	u_{tot}	$U (k = 2)$	HAC SE
α_T (K^{-1})	-8.8474×10^{-7}	1.41×10^{-9}	1.03×10^{-7}	5.97×10^{-8}	5.98×10^{-8}	1.20×10^{-7}	1.41×10^{-9}
α_H ($\% \text{RH}^{-1}$)	-1.3152×10^{-8}	1.15×10^{-9}	3.99×10^{-9}	2.30×10^{-9}	2.57×10^{-9}	5.15×10^{-9}	1.16×10^{-9}
α_P (hPa^{-1})	$+2.5949 \times 10^{-7}$	9.85×10^{-10}	1.23×10^{-8}	7.11×10^{-9}	7.17×10^{-9}	1.43×10^{-8}	9.69×10^{-10}

conditions (Sec. ??) likewise shows no enhancement over the broadband model, and because the sensor gain uncertainty (Table ??) nearly spans the observed humidity difference, we do not attribute the humidity-coefficient difference to a wavelength-specific physical effect; it is reported as an unresolved systematic pending independent humidity sensor calibration.

The significance of the humidity difference can be quantified in two ways that answer different questions. The endpoint sensitivity (Table ??) is 3.99×10^{-9} for the full worst-case humidity gain bound: the full campaign difference $\Delta\alpha_H = +4.33 \times 10^{-9}$ exceeds this endpoint by only 0.34×10^{-9} , i.e. by 0.26 of the paired bootstrap standard error of the difference ($u_{\text{stat}} = 1.30 \times 10^{-9}$). Under the rectangular gain-distribution treatment, the systematic standard uncertainty is $u_{\text{syst}} = 2.30 \times 10^{-9}$, giving a combined uncertainty $u_{\text{comb}} = \sqrt{1.30^2 + 2.30^2} \times 10^{-9} = 2.64 \times 10^{-9}$; the observed difference is then $1.64 u_{\text{comb}}$. The conclusion is robust to either treatment: the humidity difference is nearly spanned by the sensor gain sensitivity and is not attributable to a wavelength-specific physical effect with the present instrumentation.

A blocked out-of-sample validation complements the coefficient comparison: the data are divided into contiguous time blocks of 156 samples and five folds; within each fold the linear surrogate is fitted on the training blocks (including the intercept) and evaluated on the held-out blocks. Identical folds are used for both models and no global preprocessing is applied before splitting. The out-of-sample root-mean-square residual of the data surrogate is $(1.84 \pm 0.02) \times 10^{-7}$, whereas that of the Mathar surrogate is $(4.4 \pm 0.2) \times 10^{-8}$. The larger residual of the data surrogate reflects measurement noise that is absent in the analytic Mathar function; the small surrogate residual confirms that the linear model captures the Mathar environmental response to high accuracy over the observed parameter range.

IV. PHASE BUDGET FOR A REPRESENTATIVE FREE-SPACE PATH

For a representative 1 m free-space path at 1762 nm, the accumulated phase error under diurnal environmental swings of amplitude $\Delta T = 1^\circ \text{C}$, $\Delta H = 12\%$, and $\Delta P = 5 \text{ hPa}$ is computed as

$$\Delta\phi = \sqrt{(\alpha_\phi^T \Delta T)^2 + (\alpha_\phi^H \Delta H)^2 + (\alpha_\phi^P \Delta P)^2}, \quad (1)$$

where $\alpha_\phi^X \equiv \partial\phi/\partial X$ per meter (Table II of the main text). The per-meter phase budget is summarized in Table ??.

Feed-forward phase compensation uses real-time environmental sensor data to compute and cancel the predicted phase shift before each gate operation. The residual phase error after compensation has two sources: the predictive accuracy of the linear model, characterized by the regression residual standard deviation $\sigma_n = 1.84 \times 10^{-7}$, and the uncertainty in the coefficient estimates themselves, which propagates through the environmental swings ΔX_j via the coefficient standard errors σ_{α_j} . The combined residual is

$$\delta\phi_{\text{res}} = \frac{2\pi L}{\lambda} \sqrt{\sigma_n^2 + (\sigma_{\alpha_T} \Delta T)^2 + (\sigma_{\alpha_H} \Delta H)^2 + (\sigma_{\alpha_P} \Delta P)^2}. \quad (2)$$

Using the statistical (bootstrap) coefficient standard errors of Table ?? and the diurnal swings of Table ??, the coefficient-uncertainty term contributes 1.5×10^{-8} in refractive-index units, an order of magnitude below σ_n . The residual is therefore dominated by the predictive accuracy of the model, and the combined expression evaluates to $\delta\phi_{\text{res}} = 0.66 \text{ rad per meter}$.

For a fair model comparison, the static offset between the Mathar prediction and the measured refractive index must be removed, since any deployed system removes this offset with a single calibration. The static offset over the campaign is $\delta n = 1.166 \times 10^{-6}$, corresponding to 4.16 rad per meter (17.46 rad over the full 4.2 m round-trip path). After subtracting this offset, the Mathar residual standard deviation is 1.93×10^{-7} (0.69 rad per meter), compared with 1.84×10^{-7} (0.66 rad per meter) for the empirical model. The improvement in residual amplitude is therefore $I_{\text{amp}} = 4.86\%$ (95% interval [3.9, 6.0]%), and the improvement in residual variance is $I_{\text{var}} = 9.5\%$ (95% interval [7.7, 11.5]%). These modest improvements reflect that the empirical model and the debiased Mathar model agree closely over the observed parameter range, consistent with the same-condition coefficient comparison of Sec. ??.

TABLE IV. Same-condition comparison of the measured environmental coefficients with a linear surrogate of the Mathar model fitted on identical data rows. $\Delta\alpha = \alpha^{\text{data}} - \alpha^{\text{Mathar-surrogate}}$; intervals are 95% bootstrap percentile intervals.

Analysis	N	$\Delta\alpha_T$	$\Delta\alpha_H$	$\Delta\alpha_P$
In-domain (10–25 °C)	12,134	-1.41×10^{-8}	$+4.62 \times 10^{-9}$	$+3.79 \times 10^{-9}$
95% interval		$[-5.51, +2.61] \times 10^{-8}$	$[-6.86, +21.5] \times 10^{-9}$	$[+0.10, +8.05] \times 10^{-9}$
Full campaign	145,784	-3.11×10^{-9}	$+4.33 \times 10^{-9}$	$+2.39 \times 10^{-9}$
95% interval		$[-6.26, -0.15] \times 10^{-9}$	$[+1.79, +6.88] \times 10^{-9}$	$[+0.31, +4.45] \times 10^{-9}$

TABLE V. Phase error budget for a 1 m free-space optical path at 1762 nm under typical diurnal environmental swings. Sensitivities are from Table II of the main text.

Parameter	ΔX (diurnal)	α_ϕ^X	$\Delta\phi$ (rad)
Temperature	1 °C	$3.16 \text{ rad m}^{-1} \text{ K}^{-1}$	3.16
Relative humidity	12 %	$0.047 \text{ rad m}^{-1} (\% \text{RH})^{-1}$	0.56
Pressure	5 hPa	$0.93 \text{ rad m}^{-1} \text{ hPa}^{-1}$	4.63
Total (RSS)			5.63

V. MODEL LIMITATIONS AND SCOPE OF THE MEASUREMENT APPROACH

Several factors set the achievable accuracy of the empirical model. First, the model assumes linear, decoupled environmental dependences; while cross-correlations among T , H , and P are small in our well-ventilated measurement corridor, a full multivariate polynomial including cross-terms could further reduce σ_n toward the 10^{-8} level, as suggested by the small nonlinearity component of the residual (Table ??). Second, CO_2 variations are not tracked in the present setup; seasonal CO_2 changes of 15 ppm to 25 ppm alter n by approximately 3×10^{-8} , which lies below the current residual standard deviation but may become relevant as sensor performance improves.

Beyond these model-level uncertainties, the measurement approach itself has inherent scope limitations that define what the calibrated coefficients can and cannot correct in a deployed system. (i) *Local gradients*: the environmental sensor samples temperature, humidity, and pressure at a single location, whereas the optical path extends over several meters. Under strong thermal stratification or localized humidity sources, the sensor reading may deviate from the path-averaged value, introducing an uncompensated residual that scales with the gradient magnitude. This is particularly relevant for vertical beam paths or links traversing multiple thermal zones. (ii) *Turbulence*: the sensor sampling rate (2 Hz) is far slower than the characteristic timescales of turbulent refractive-index fluctuations (milliseconds to seconds) [?]. The feed-forward scheme therefore corrects only the slowly varying background but not the stochastic Kolmogorov-phase noise, which must be addressed by other means such as high-bandwidth phase tracking or spatial mode filtering. (iii) *Unsensed geometric path-length variations*: the dual-wavelength common-mode rejection cancels mechanical fluctuations only to the extent that both wavelengths experience identical geometric displacement. Differential effects, such as chromatic beam wander due to thermal

lensing in transmissive optics or wavelength-dependent thermal expansion of mechanical mounts, produce an uncompensated residual that is not sensed by the environmental monitor. (iv) *Fast mechanical noise*: the interferometer rejects common-mode vibrations within its fringe-tracking bandwidth, but spectral components above this bandwidth, or differential vibrations between the two wavelength channels due to spatially separated beam paths on the detection side, are not canceled and set a practical noise floor for the phase correction. These scope limitations are inherent to any single-point, slow-sampling feed-forward approach and apply equally to the empirical and the Mathar-based compensation.

VI. KRAMERS-KRONIG ANALYSIS OF THE HUMIDITY COEFFICIENT

The proximity of the probe wavelength to water-vapor absorption features (Sec. III of the main text) motivates a spectroscopic estimate of the humidity-coefficient contribution. No environmental time-series data enter this calculation; it is a purely spectroscopic estimate intended to test whether the sign and order of magnitude of the measured humidity response are reproduced by the known water-vapor line list.

Kramers-Kronig integration. The refractive-index change per unit humidity at frequency ν follows from the absorption spectrum $\kappa(\nu')$ via the principal-value integral [?]

$$\Delta n(\nu) = \frac{1}{\pi} \mathcal{P} \int \frac{\kappa(\nu')}{\nu' - \nu} d\nu'. \quad (3)$$

The absorption spectrum is constructed from H_2O line positions, strengths, and broadening parameters in the HITRAN2020 database [?], using a Lorentzian line shape. The water-vapor number density per 1% RH is obtained from the Magnus saturation-vapor-pressure formula. The integral excludes a narrow interval $\pm \varepsilon$ around

ν to handle the singularity; convergence with respect to ε was verified.

Sensitivity to integration range and atmospheric parameters. The Kramers-Kronig prediction depends weakly on the integration range (varying by about 1% between 1600–1950 nm and 1500–2050 nm) and noticeably on the assumed atmospheric conditions through the water-vapor number density. Table ?? summarises the prediction for four combinations of integration range and atmospheric parameters. The first two rows use the mean conditions of the campaign ($T = 302.0$ K, $P = 986.4$ hPa, corresponding to the measured mean temperature of 28.9°C); the last two rows use the warmer reference conditions of 303 K and 990 hPa for comparison with earlier iterations of this analysis.

TABLE VI. Sensitivity of the Kramers-Kronig humidity-coefficient prediction to integration range and atmospheric parameters. All integrations use a quadrature subinterval limit of 2000, verified to converge (relative error below 1%). The measured α_H is $-1.3152 \times 10^{-8} \text{ \%RH}^{-1}$; the Mathar surrogate fitted on identical data rows (Sec. ??) is $-1.7486 \times 10^{-8} \text{ \%RH}^{-1}$.

Integration range	T (K)	P (hPa)	α_H^{KK} ($\text{\%RH}^{-1}$)
1600–1950 nm	302	986	-1.3500×10^{-8}
1500–2050 nm	302	986	-1.3582×10^{-8}
1600–1950 nm	303	990	-1.4361×10^{-8}
1500–2050 nm	303	990	-1.4438×10^{-8}

Interpretation. The Kramers-Kronig prediction consistently yields a negative humidity coefficient of order $10^{-8} \text{ \%RH}^{-1}$, confirming that the sign and rough magnitude of the measured humidity response are compatible with water-vapor dispersion. At the mean experimental conditions the prediction is $-1.35 \times 10^{-8} \text{ \%RH}^{-1}$ (1600–1950 nm range), close to the measured coefficient ($-1.32 \times 10^{-8} \text{ \%RH}^{-1}$, agreement within 3%) and smaller in magnitude than the Mathar surrogate ($-1.75 \times 10^{-8} \text{ \%RH}^{-1}$). The line-by-line HITRAN2020 water-vapor spectrum therefore does not predict an enhancement of the humidity response over the broadband model at 1762 nm. This is consistent with the same-condition comparison of Sec. ??, where the measured humidity coefficient is found to be smaller in magnitude than the Mathar surrogate rather than enhanced. The remaining difference between the measured coefficient and the spectroscopic prediction is comparable to or smaller than the sensor gain systematic (Table ??); given this unresolved systematic, a quantitative attribution of the residual difference to a wavelength-specific physical effect is not possible with the present data.